

termvar, x
label, l, k *field label*
index, i, j, n, m *indices*

terminals	$::=$ $\begin{array}{ l} \longrightarrow \\ \rightarrow \\ \Rightarrow \\ \lambda \\ \mapsto \\ \vdash \\ \emptyset \\ \times \\ <: \\ \langle \\ \rangle \end{array}$		
Γ	$::=$ $\begin{array}{ l} \emptyset \\ \Gamma, x : T \end{array}$		<i>contexts:</i> <i>empty context</i> <i>term variable binding</i>
labels	$::=$ $\begin{array}{ l} l_1 .. l_n \end{array}$		
T, S, U	$::=$ $\begin{array}{ l} T \rightarrow T' \\ (T) \\ \text{Top} \\ \{ l_i : T_i \}^{i \in 1..n} \end{array}$	M	<i>types:</i> <i>type of functions</i> <i>maximum type</i> <i>type of records</i>
t	$::=$ $\begin{array}{ l} (t) \\ x \\ \lambda x : T. t \\ t_1 t_2 \\ [x \mapsto t'] t \\ \{ l_i = t_i \}^{i \in 1..n} \\ t.l \end{array}$	M bind x in t M	<i>terms:</i> <i>variable</i> <i>abstraction</i> <i>application</i> <i>record</i> <i>projection</i>
v	$::=$ $\begin{array}{ l} \lambda x : T. t \\ \{ l_i = v_i \}^{i \in 1..n} \end{array}$	bind x in t	<i>values:</i> <i>abstraction value</i> <i>record value</i>
formula	$::=$ $\begin{array}{ l} \text{judgement} \\ x : T \in \Gamma \\ \text{DISTINCT } l_1 .. l_n \\ \text{formula}_1 .. \text{formula}_n \\ j \text{ INDEXES } t_1 .. t_n \\ j \text{ INDEXES } T_1 .. T_n \\ T_1 \text{ is a permutation of } T_2 \end{array}$		
Jop	$::=$		

		$t \longrightarrow t'$	<i>Evaluation</i>
Jtype	::=		
		$\Gamma \vdash t : T$	<i>Typing</i>
		$T_1 <: T_2$	<i>Subtyping</i>
judgement	::=		
		Jop	
		Jtype	
user_syntax	::=		
		termvar	
		label	
		<i>index</i>	
		terminals	
		Γ	
		labels	
		T	
		t	
		v	
		formula	

$\boxed{t \longrightarrow t'}$ *Evaluation*

$$\begin{array}{c}
\frac{t_1 \longrightarrow t'_1}{t_1 t_2 \longrightarrow t'_1 t_2} \quad \text{E_APP1} \\
\frac{t_2 \longrightarrow t'_2}{v_1 t_2 \longrightarrow v_1 t'_2} \quad \text{E_APP2} \\
\frac{}{(\lambda x : T. t_{12}) v_2 \longrightarrow [x \mapsto v_2] t_{12}} \quad \text{E_APPABS} \\
\frac{}{\{l_i = v_i^{i \in 1..n}\}.l_j \longrightarrow v_j} \quad \text{E_PROJRCD} \\
\frac{t_1 \longrightarrow t'_1}{t_1.l \longrightarrow t'_1.l} \quad \text{E_PROJ} \\
\frac{t \longrightarrow t'}{\{l_i = v_i^{i \in 1..m}, l = t, l'_j = t_j^{j \in 1..n}\} \longrightarrow \{l_i = v_i^{i \in 1..m}, l = t', l'_j = t_j^{j \in 1..n}\}} \quad \text{E_RCD}
\end{array}$$

$\boxed{\Gamma \vdash t : T}$ *Typing*

$$\begin{array}{c}
\frac{x : T \in \Gamma}{\Gamma \vdash x : T} \quad \text{T_VAR} \\
\frac{\Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \lambda x : T_1. t_2 : T_1 \rightarrow T_2} \quad \text{T_ABS} \\
\frac{\Gamma \vdash t_1 : T_{11} \rightarrow T_{12} \quad \Gamma \vdash t_2 : T_{11}}{\Gamma \vdash t_1 t_2 : T_{12}} \quad \text{T_APP} \\
\frac{\Gamma \vdash t : S \quad S <: T}{\Gamma \vdash t : T} \quad \text{T_SUB}
\end{array}$$

$$\begin{array}{c}
\frac{\Gamma \vdash \mathbf{t}_i : \mathbf{T}_i^{i \in 1..n} \quad \text{DISTINCT } \mathbf{l}_i^{i \in 1..n}}{\Gamma \vdash \{ \mathbf{l}_i = \mathbf{t}_i^{i \in 1..n} \} : \{ \mathbf{l}_i : \mathbf{T}_i^{i \in 1..n} \}} \quad \text{T_RCD} \\
\\
\frac{\Gamma \vdash \mathbf{t}_1 : \{ \mathbf{l}_i : \mathbf{T}_i^{i \in 1..n} \}}{\Gamma \vdash \mathbf{t}_1.\mathbf{l}_j : \mathbf{T}_j} \quad \text{T_PROJ}
\end{array}$$

$\boxed{\mathbf{T}_1 <: \mathbf{T}_2}$ *Subtyping*

$$\begin{array}{c}
\overline{\mathbf{S} <: \mathbf{S}} \quad \text{S_REFL} \\
\\
\frac{\mathbf{S} <: \mathbf{U} \quad \mathbf{U} <: \mathbf{T}}{\mathbf{S} <: \mathbf{T}} \quad \text{S_TRANS} \\
\\
\frac{\mathbf{T}_1 <: \mathbf{S}_1 \quad \mathbf{S}_2 <: \mathbf{T}_2}{\mathbf{S}_1 \rightarrow \mathbf{S}_2 <: \mathbf{T}_1 \rightarrow \mathbf{T}_2} \quad \text{S_ARROW} \\
\\
\overline{\mathbf{S} <: \text{Top}} \quad \text{S_TOP} \\
\\
\overline{\{ \mathbf{l}_i : \mathbf{T}_i^{i \in 1..m}, \mathbf{l}'_j : \mathbf{T}'_j^{j \in 1..n} \} <: \{ \mathbf{l}_i : \mathbf{T}_i^{i \in 1..m} \}} \quad \text{S_RCDWIDTH} \\
\\
\frac{\mathbf{S}_i <: \mathbf{T}_i^{i \in 1..m}}{\{ \mathbf{l}_i : \mathbf{S}_i^{i \in 1..m} \} <: \{ \mathbf{l}_i : \mathbf{T}_i^{i \in 1..m} \}} \quad \text{S_RCDDEPTH} \\
\\
\frac{\{ \mathbf{k}_j : \mathbf{S}_j^{j \in 1..n} \} \text{ is a permutation of } \{ \mathbf{l}_i : \mathbf{T}_i^{i \in 1..n} \}}{\{ \mathbf{k}_j : \mathbf{S}_j^{j \in 1..n} \} <: \{ \mathbf{l}_i : \mathbf{T}_i^{i \in 1..n} \}} \quad \text{S_RCDPERM}
\end{array}$$

Definition rules: 19 good 0 bad
Definition rule clauses: 38 good 0 bad